Boeing Mishaps in 2024

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1. Boeing Accidents in 2024

Intro of Boeing's 4 Incidents in 2024

Alaska 1282 Door-plug

Door-plug panel separated mid-flight due to missing bolts that were never reattached after prior rework; rapid depressurization; injuries minor

Jan 5, 2024

Delta Nose-gear wheel

Nose-gear wheel separated during lineup for takeoff; no injuries

Jan 20, 2024

United Main-gear tire

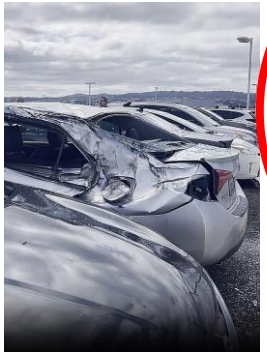
Main-gear tire separated after takeoff; safe diversion; ground property damage

Mar 7, 2024

Southwest Engine cowling

Engine cowling detached on takeoff and struck wing flap; returned safely

Apr 7, 2024



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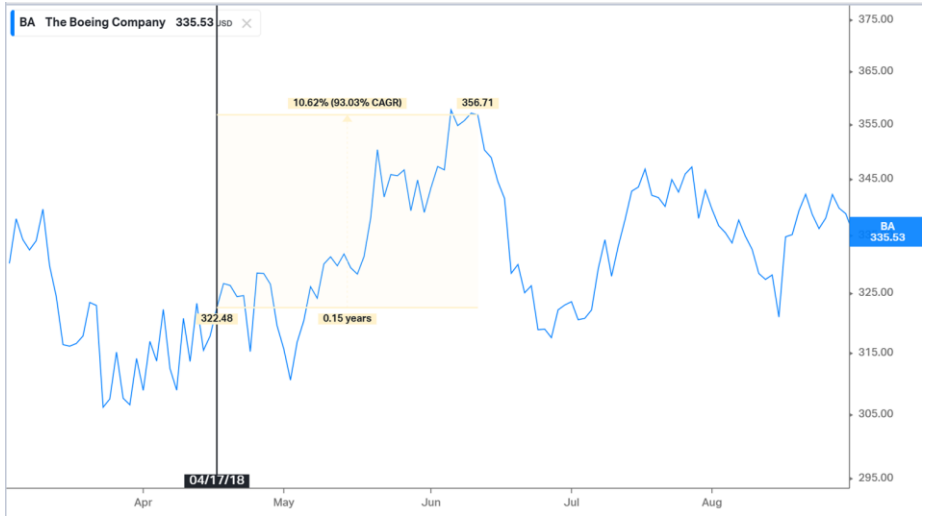
Similar Cases

1. Southwest Airlines Flight 1380 (2018 — **Boeing 737-700**): Engine fan blade on a CFM56-7B engine fractured mid-flight, causing an uncontained failure and one fatality.

2. British Airways Flight 2276 (2015 — **Boeing 777-200ER**): Uncontained GE90 engine failure during takeoff at Las Vegas; aircraft evacuated safely.

3. British Airways Flight 38 (2008 — **Boeing 777-200ER**): Aircraft lost thrust on both Rolls-Royce Trent 800 engines during approach to Heathrow.



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2. Why Different? – MCAS

Lion Air, Ethiopian Airlines

1. Lion Air Penerbangan 610:
    - Event Summary: The plane hit the sea at high speed (≈700 km/h), breaking apart instantly and disintegrating on impact, leaving no survivors.
1. Ethiopian Airlines Flight 302
    - Event Summary: The jet crashed near Bishoftu, Ethiopia, at high speed, creating a crater and scattering debris with no survivors.

Similarity: A faulty angle-of-attack sensor sent incorrect data, falsely indicating the aircraft's nose was too high. This triggered the **MCAS system**, which repeatedly **forced the nose down**. The pilots struggled to counter it, but the system kept reactivating, leading to loss of control and a high-speed dive that destroyed the aircraft on impact.

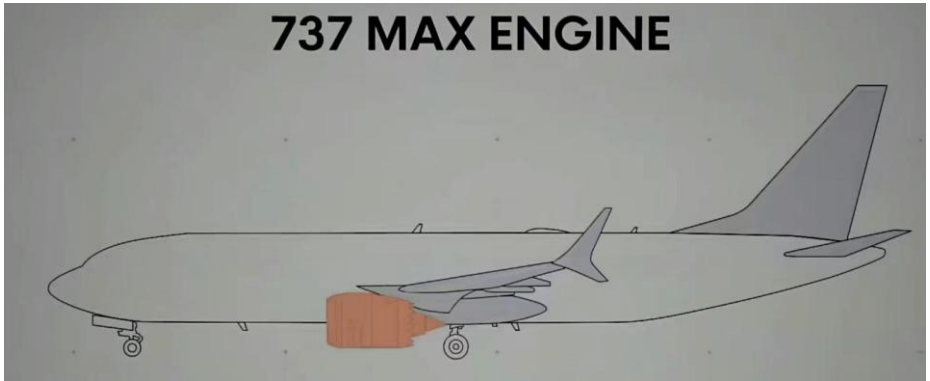
What is MCAS system? Why was the MCAS system designed?

Why Did Outstanding Boeing Design Poor MCAS

- At the beginning, Boeing focused on **quality, innovation, and safety**. However, after acquiring McDonnell Douglas on August 1, 1997, Boeing's direction changed. The former McDonnell Douglas CEO became Boeing's president and COO, and the company's goal shifted from building industry-leading aircraft to maintaining **stock prices** and producing good **financial reports**.

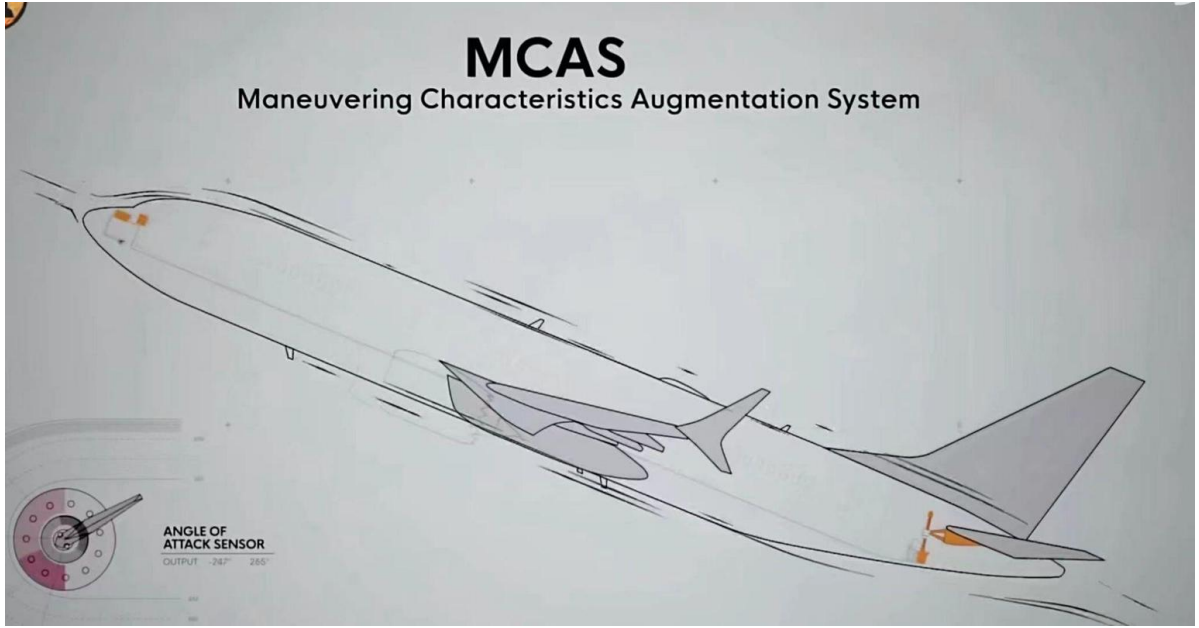
- In 2010, Airbus introduced the A320neo, which offered 15% better fuel efficiency, causing panic at Boeing. Having long neglected investment in new aircraft development, Boeing could only update the 737NG instead of designing a new model, resulting in the 737 MAX. Boeing promised that **no flight simulator training** would be required and that the MAX would be even more fuel-efficient than the A320neo.

- However, the larger engines on the 737 MAX caused the aircraft to pitch upward excessively during flight, reducing lift and increasing the risk of a stall.



MCAS (Maneuvering Characteristics Augmentation System)

- Essence of MCAS: **prevent stalls**. When sensors detected that the aircraft's nose was pitching up excessively, the software automatically adjusted the **horizontal stabilizer** to push the nose down, assisting the pilots in maintaining control.



MCAS (Continue)

- However, according to regulations, adding a new system to the 737 would have required additional pilot training and certification. Because Boeing had **promised that no simulator training** would be needed and was eager to bring the 737 MAX to market quickly, the company **misrepresented** MCAS as merely an "**addition to the speed trim system**" rather than a new safety-critical feature.

- Originally, MCAS was not designed to override pilot input, since excessive nose-up situations were expected to occur only at high speeds. But during testing, similar conditions appeared even at lower speeds, prompting Boeing to expand MCAS's operating range and increase its authority—allowing it to **override pilot commands**.

- design flaw: although the aircraft had **two** angle-of-attack (AoA) sensors, MCAS relied on input from **only one**. If that single sensor failed, the system could activate erroneously, with catastrophic consequences.

Boeing's Response After the Two Crashes

- Lion Airline (Indonesia):
  - On the day of crash, the weather was clear, the aircraft was new, and Lion Air already had a poor safety reputation. As a result, few people initially suspected Boeing, and blame quickly shifted toward the airline and its pilots.
  - However, investigations later revealed deeper issues. A Federal Aviation Administration (FAA) risk analysis had warned that, without changes, the 737 MAX could become one of the most dangerous modern airliners. Indonesian investigators also found signs of an unusual force continually pushing the aircraft's nose downward during the flight.
  - Boeing later confirmed that the MCAS had been activated—a system that was not described in the flight manual and unknown to the pilots. The company argued that the crew failed to disable MCAS, while investigators pointed out that the pilots had never been trained to handle it.

Boeing then promised to **fix the MCAS software within six weeks**, while still claiming it was **uncertain whether MCAS alone had caused the crash**.

- Five Months Later: Ethiopian Airlines
  - Boeing's CEO repeatedly assured the public that the aircraft was safe, again shifting blame onto the pilots.

Risk Analysis

1. Operational Risk

- Pilot training and information gaps:
  Crews were not fully informed of MCAS's existence or trained on how to respond to its malfunctions.
- Maintenance and calibration issues:
  Faulty angle-of-attack (AoA) sensors and inconsistent maintenance practices directly contributed to the accidents.
- Communication breakdowns:
  Critical safety information was not properly shared between engineering, training, and airline customers.

2. Model Risk (Design & Engineering Risk)

- Single-sensor dependency:
  MCAS relied on input from only one AoA sensor, creating a single point of failure.
- System authority imbalance:
  MCAS could repeatedly override pilot inputs—an unacceptable control hierarchy flaw.
- Software validation risk:
  Inadequate testing and assumptions that "pilots will manage it" underestimated real-world behavior.

Risk Analysis (Continue)

3. Regulatory / Compliance Risk

- Self-certification conflict:
  The FAA delegated excessive oversight authority to Boeing (acting as both “player and referee”), weakening independent review.
- Regulatory capture:
  Close ties between Boeing and regulators led to lenient certification standards and insufficient scrutiny.
- Documentation and transparency risk:
  Boeing’s omission of MCAS details in flight manuals and training materials constituted non-disclosure of safety-critical information.

Result

- After the Ethiopian Airlines crash, China became the first country to ground all Boeing 737 MAX aircraft, prompting many other nations to follow and issue similar grounding orders.

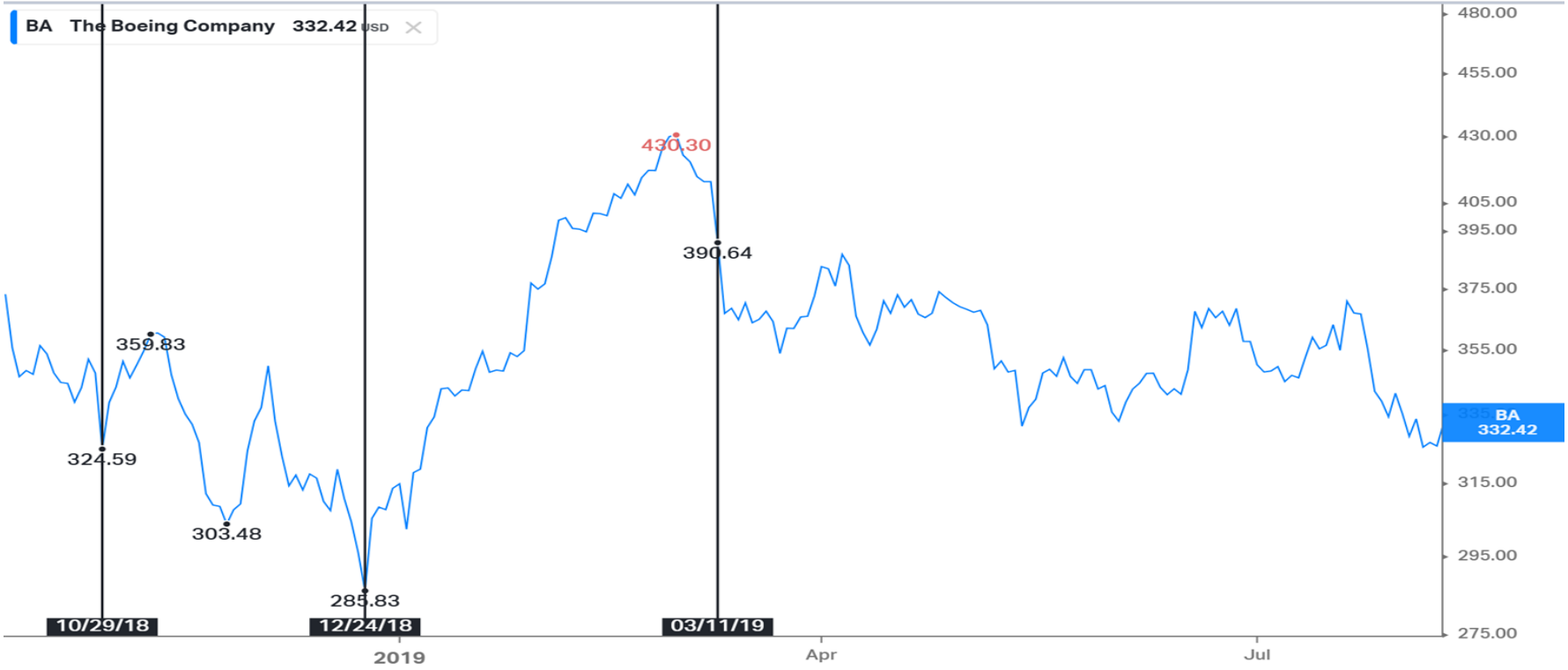
- Ethiopia sent the flight's black boxes to France, rather than the United States, for data extraction and analysis — a decision that reflected diminished trust in Boeing and U.S. regulatory authorities.

- The 737 MAX became the only aircraft in history to be grounded by direct order of a U.S. president. Following the grounding, all 737 MAX aircraft had to be flown empty to designated storage sites, yet many pilots refused to operate these transfer flights.

- Boeing's stock price fell

3. Similar Cases, Different Outcomes

Finance: Lion & Ethiopia



Finance: After MCAS (2024)

4. Conclusion

Why Different? – Trust risk

- Power of Trust:
  Trust is the foundation of sustainable business. It connects culture, operations, reputation, and leadership. Once trust is lost, rebuilding it takes far more time and cost than any short-term gain can justify.
- Culture–Operation–Reputation–Trust–Trust Management:
  A company's internal culture shapes its daily operations. Operations build its reputation. Reputation determines the level of trust from customers, regulators, and investors. Managing this trust chain is therefore essential for long-term success.
- manage trust risk:
  Trust risk arises when actions, decisions, or communication damage stakeholders' confidence. It must be managed just like financial or operational risk — through transparency, accountability, and ethical leadership.
- How to manage trust risk
  For Boeing, managing trust risk means shifting focus from stock prices and revenue to safety and quality. Only by prioritizing integrity and engineering excellence can the company rebuild public confidence. In the long run, trust becomes the driver of shareholder value and revenue, not the obstacle.

Thank you.